

DATA ANALYTICS INTERNSHIP

-InternSpark

(AICTE-Listed Internship Program)

REPORT ON

SALES DATA ANALYSIS REPORT

(TASK 2)

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Introduction

Sales data analysis plays an important role in understanding business performance, customer behavior, and market trends. The objective of this project is to analyze the Superstore Sales dataset and extract meaningful insights that can help improve business decisions.

The analysis focuses on identifying sales trends, regional performance, product demand, shipping preferences, and seasonal patterns. Various data visualization techniques have been used to present the findings in an easy-to-understand manner.

Objective

The primary objectives of this project are:

- To clean and preprocess the sales dataset.
- To calculate important business metrics.
- To identify monthly sales trends.
- To analyze regional sales performance.
- To determine the best-selling products.
- To understand customer shipping preferences.
- To provide actionable business recommendations.

Dataset Overview

The dataset contains transactional sales information including:

- Order ID
- Order Date
- Product Name
- Category
- Sub-Category
- Region
- Sales Amount
- Shipping Mode

This data helps in evaluating overall business performance and identifying opportunities for growth.

Data Cleaning and Preprocessing:-

Before performing the analysis, the dataset was cleaned to ensure accuracy and reliability.

The following preprocessing steps were performed:

Handling Missing Values

Rows containing missing values in Sales and Order_Date columns were removed.

Date Conversion

The Order_Date column was converted into datetime format for time-series analysis.

Duplicate Removal

Duplicate records were identified and removed to avoid incorrect results.

Invalid Data Removal

Records with zero or negative sales values were removed from the dataset.

After cleaning, the dataset became suitable for analysis and visualization.

Key Performance Metrics

The following key business metrics were calculated:-

Total Revenue

Total revenue represents the overall sales generated by the business.

Total Orders

The total number of unique orders placed by customers.

Average Order Value (AOV)

Average Order Value indicates the average amount spent per order and is calculated as:

$$\text{AOV} = \text{Total Revenue} / \text{Total Orders}$$

These metrics provide a quick overview of business performance.

Data Visualization and Analysis:-

Monthly Sales Trend

A line chart was created to analyze monthly sales performance.

Insights:

- Sales fluctuated across different months.
- Certain months generated significantly higher revenue than others.

- Seasonal patterns can be observed, indicating opportunities for targeted marketing campaigns.

(Insert Monthly Sales Trend Graph Here)

Sales by Region

A bar chart was used to compare sales across different regions.

Insights:

- Some regions contributed more revenue than others.
- High-performing regions can be targeted for further expansion.
- Low-performing regions may require additional marketing efforts.

(Insert Sales by Region Graph Here)

Monthly Sales Seasonality

A boxplot was created to study sales distribution across different months.

Insights:

- Certain months exhibited higher median sales.
- Seasonal demand variations were clearly visible.
- Understanding seasonality can help improve inventory planning.

(Insert Monthly Sales Seasonality Graph Here)

Top 10 Products by Sales

The top-selling products were identified based on total sales revenue.

Insights:

- A small number of products generated a significant portion of overall revenue.
- These products should receive priority in marketing and inventory management.
- High-performing products can be used for cross-selling opportunities.

(Insert Top 10 Products Graph Here)

Shipping Mode Preference

A count plot was used to analyze customer preferences for shipping methods.

Insights:

- Customers showed clear preferences for specific shipping modes.
- Popular shipping methods should be optimized for better customer satisfaction.
- Faster and reliable delivery can improve customer retention.

(Insert Shipping Mode Preference Graph Here)

Sales Heatmap: Category vs Sub-Category

A heatmap was created to visualize sales distribution across categories and sub-categories.

Insights:

- Certain sub-categories generated substantially higher sales.
- Underperforming categories can be reviewed for improvement.
- Product portfolio optimization can be based on these findings.

(Insert Heatmap Here)

Tools Used:

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Google Colab

Key Findings

The analysis revealed several important business insights:

- Sales performance varies significantly throughout the year.
- Some regions contribute a larger share of revenue.
- A few products are responsible for a major portion of total sales.
- Customers prefer specific shipping methods.
- Seasonal trends affect sales patterns.
- Product categories show varying levels of profitability and demand.

These findings provide valuable information for strategic decision-making.

Recommendations for Alfido Tech

Based on the analysis, the following tactical recommendations are proposed:

1. Focus on High-Performing Products

Increase promotional efforts and inventory availability for top-selling products to maximize revenue.

2. Strengthen Low-Performing Regions

Introduce targeted advertising campaigns, regional discounts, and customer engagement initiatives in regions with lower sales.

3. Optimize Shipping Operations

Improve delivery efficiency for the most preferred shipping methods to enhance customer satisfaction and retention.

4. Launch Seasonal Marketing Campaigns

Use sales seasonality insights to create promotional campaigns during peak demand periods and boost sales during slower months.

5. Implement Cross-Selling and Upselling Strategies

Recommend complementary products and premium alternatives to customers to increase average order value.

Conclusion

This project successfully analyzed the Superstore Sales dataset using Python and various data visualization techniques. The analysis provided meaningful insights into sales trends, customer preferences, product performance, and regional sales distribution.

The findings indicate that focusing on high-demand products, improving low-performing regions, optimizing shipping services, and implementing seasonal marketing strategies can significantly enhance business performance. These data-driven recommendations can help Alfido Tech improve customer satisfaction, increase revenue, and achieve sustainable business growth.

Overall, the project demonstrates how data analytics can support informed decision-making and contribute to organizational success.

References

- Python Documentation
- Pandas Documentation

- NumPy Documentation
- Matplotlib Documentation
- Seaborn Documentation
- Superstore Sales Dataset

Screenshots and Visuluzation:-

Click to add a breakpoint

```

import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
df=pd.read_csv('Superstore_Sales.csv', encoding='latin1')
print(df.head())
print(df.info())

```

	Row_ID	Order_ID	Order_Date	Ship_Date	Ship_Mode	Customer_ID	\
0	1	CA-2017-152156	8/11/2017	11/11/2017	Second Class	CG-12520	
1	2	CA-2017-152156	8/11/2017	11/11/2017	Second Class	CG-12520	
2	3	CA-2017-138688	12/6/2017	16/06/2017	Second Class	DV-13045	
3	4	US-2016-108966	11/10/2016	18/10/2016	Standard Class	SO-20335	
4	5	US-2016-108966	11/10/2016	18/10/2016	Standard Class	SO-20335	

	Customer_Name	Segment	Country	City	State	\
0	Claire Gute	Consumer	United States	Henderson	Kentucky	
1	Claire Gute	Consumer	United States	Henderson	Kentucky	
2	Darrin Van Huff	Corporate	United States	Los Angeles	California	
3	Sean O Donnel	Consumer	United States	Fort Lauderdale	Florida	
4	Sean O Donnel	Consumer	United States	Fort Lauderdale	Florida	

	Postal_Code	Region	Product_ID	Category	Sub_Category	\
0	42420.0	South	FUR-BO-10001798	Furniture	Bookcases	
1	42420.0	South	FUR-CH-10000454	Furniture	Chairs	
2	90036.0	West	OFF-LA-10000240	Office Supplies	Labels	
3	33311.0	South	FUR-TA-10000577	Furniture	Tables	
4	33311.0	South	OFF-ST-10000760	Office Supplies	Storage	

	Product_Name	Sales
0	Bush Somerset Collection Bookcase	261.9600
1	Hon Deluxe Fabric Upholstered Stacking Chairs,...	731.9400
2	Self-Adhesive Address Labels for Typewriters b...	14.6200

```
2 Self-Adhesive Address Labels for Typewriters b... 14.6200
```

```
...
```

```
17 Sales          9800 non-null    float64
```

```
dtypes: float64(2), int64(1), object(15)
```

```
memory usage: 1.3+ MB
```

```
None
```

Output is truncated. View as a [scrollable element](#) or open in a [text editor](#). Adjust cell output [settings](#)...

```
print("Missing values before cleaning:\n", df.isnull().sum())

df = df.dropna(subset=['Sales', 'Order_Date'])

df['Order_Date'] = pd.to_datetime(df['Order_Date'], dayfirst=True)

df = df.drop_duplicates()
df = df[df['Sales'] > 0]

total_revenue = df['Sales'].sum()
total_orders = df['Order_ID'].nunique()
aov = total_revenue / total_orders

print(f"Total Revenue: {total_revenue}")
print(f"Average Order Value: {aov}")
```

```
... Missing values before cleaning:
```

```
Row_ID      0
Order_ID    0
Order_Date  0
Ship_Date   0
Ship_Mode   0
```

```
Missing values before cleaning:
```

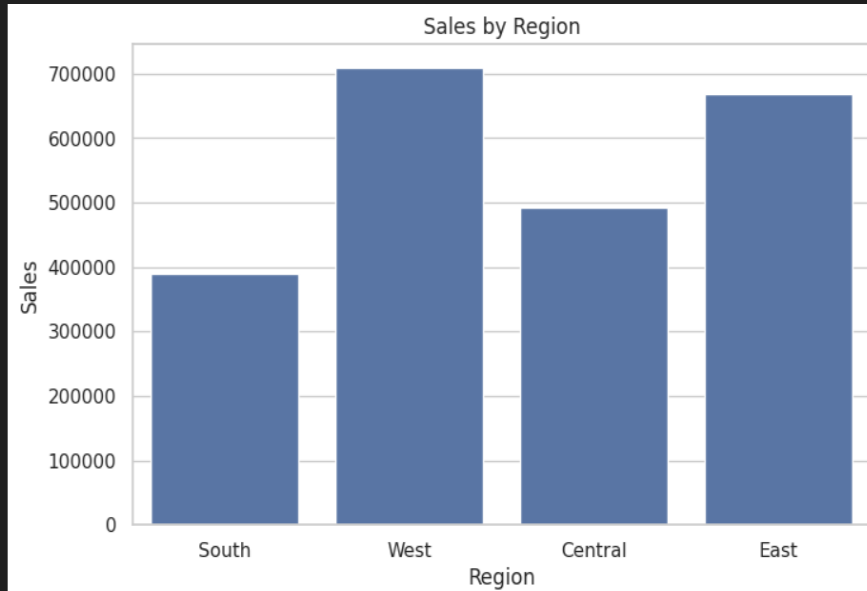
```
Row_ID      0
Order_ID    0
Order_Date  0
Ship_Date   0
Ship_Mode   0
Customer_ID 0
Customer_Name 0
Segment     0
Country     0
City        0
State       0
Postal_code 11
Region      0
Product_ID  0
Category    0
Sub_Category 0
Product_Name 0
Sales       0
dtype: int64
Total Revenue: 2261536.7827000003
Average Order Value: 459.4751691791955
```

```
import matplotlib.pyplot as plt
import seaborn as sns

sns.set_theme(style="whitegrid")

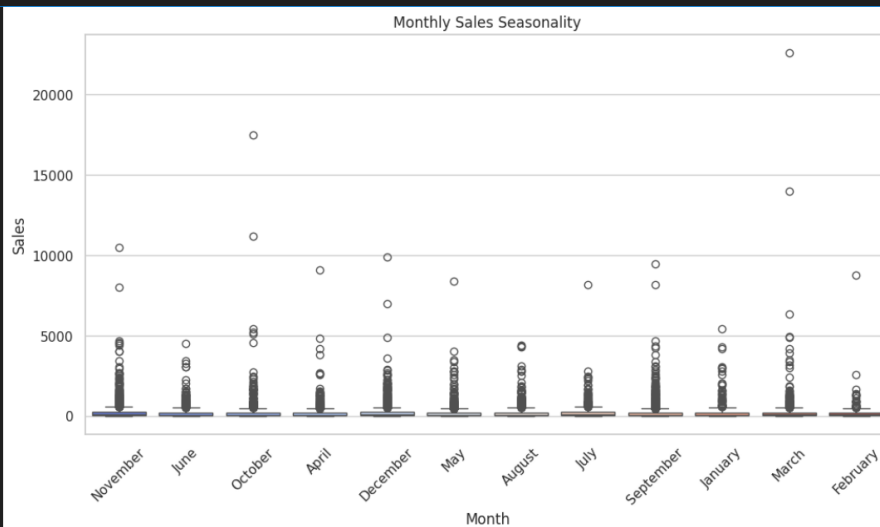
df.groupby(df['Order_Date'].dt.to_period('M'))['Sales'].sum().plot(kind='line', figsize=(10, 5), marker='o')
plt.title('Monthly Sales Trend')
```

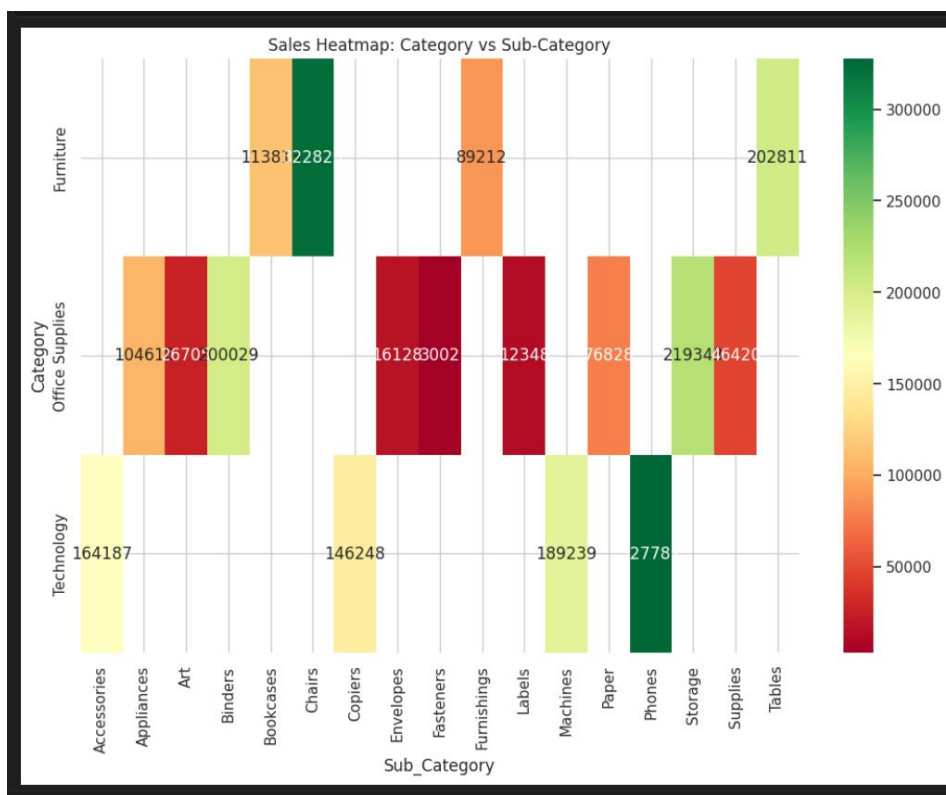
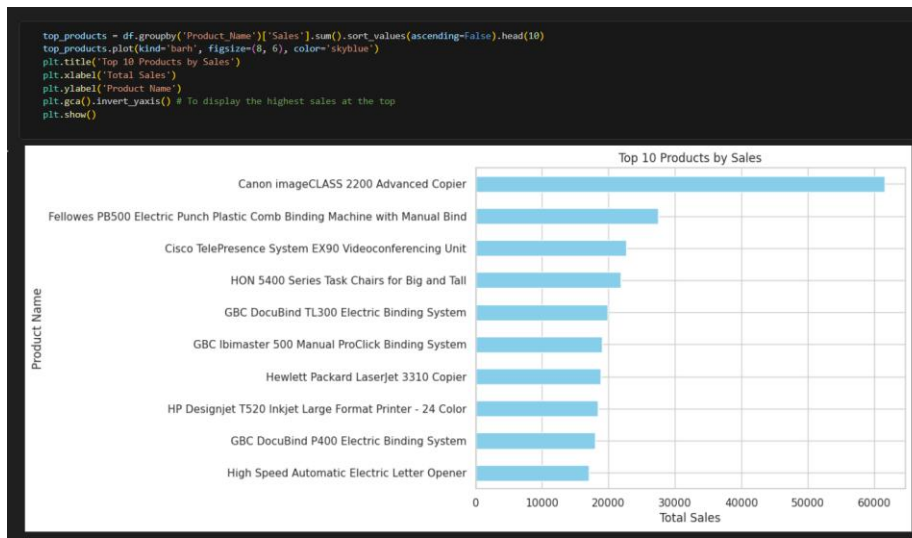
```
plt.figure(figsize=(8, 5))
sns.barplot(data=df, x='Region', y='Sales', estimator=sum, errorbar=None)
plt.title('Sales by Region')
plt.show()
```



```
df['Order_Date'] = pd.to_datetime(df['Order_Date'], dayfirst=True, errors='coerce')
df['Month'] = df['Order_Date'].dt.month_name()

plt.figure(figsize=(12,6))
sns.boxplot(data=df, x='Month', y='Sales', palette='coolwarm', hue='Month', legend=False)
plt.title('Monthly Sales Seasonality')
plt.xticks(rotation=45)
plt.show()
```





Code Link:-

<https://github.com/Shadnmnn34/InternSpark--InternShip/tree/8f411d9e841fb95cedf18149011bfb90fefadde7/Sales%20Performances%20Analysis>